

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav-senthil-kumar | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Data Science & ML (Python), Computer Science I (Java), Computer Science II (C++), Calculus III
- **Activities:** AI Alignment, Agentic AI, Disruption Lab, SIGRobotics, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the document search engine (*Python FastAPI, Elasticsearch*) for a competitive high school debate research SaaS platform hosted on *AWS*, developed by an *Agile* team of **10 engineers** and drawing **10K+ site visitors**
- Designed and built a text-retrieval algorithm that extracts evidence segments from debate cases by validating source URLs and filtering statistical outliers with **IQR** on the word-count distribution of tag, citation and evidence text
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat
- Indexed **600K+ debate-evidence documents** on *Elasticsearch* via *FastAPI*, returning results in **under 1 second**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Hootsuite*, *Power BI* and *Microsoft Excel*
- Identified that “science fun facts” posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

Product Development Intern | Jetson

January 2025 – February 2025

- Delivered a **5-page report** recommending strategies to optimize user engagement directly to Jetson’s CEO
- Redesigned user onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led team of **4** to build a pricing algorithm for **Kamion (YC S22)**. **2nd of 7 at UIUC Founder’s Hackathon**
- Scraped **15,000** truck-listing images for *data mining*, auditing quality via de-duplication and non-USD filtering
- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from photos
- Revised the schema after each wrong prediction until **10 consecutive images** were correct on each of **3 features**
- Built a *quantile regression* pricing model on the extracted features, reaching **87% accuracy** on **30 samples**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Deployed a **RAG agent** (*Claude Sonnet 4.5*) that retrieves session-scoped vector documents (*Supermemory*)
- Stored paper data on *Supabase (PostgreSQL)* and wrote *SQL* queries to filter papers by field, year and citation count
- Built a dual-API ingestion pipeline resolving papers via *OpenAlex* & backfilling metadata from *Semantic Scholar*

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* neural network (**4.39M** parameters) in *PyTorch*, trained on **15K+ images** to classify waste into 4 disposal pathways at **96.13% accuracy**. Published a paper on *arXiv* and 2 datasets on *Kaggle*
- Built a synthetic data generation pipeline using *U2-Net* segmentation and alpha-compositing to generate 9K images to train the model to detect contamination in recyclable waste. **Placed 2nd at the Arizona State Science Fair**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript

ML & Data: PyTorch, TensorFlow, Keras, scikit-learn, pandas, NumPy, OpenCV, Grad-CAM, Random Forest

GenAI: RAG, Claude API, Cursor AI SDK, Claude MCP, Gemma 4, Vercel AI SDK, Supermemory

Platforms: PostgreSQL, Supabase, Elasticsearch, Power BI, Excel, AWS, Google Cloud, Vercel

Developer Tools: Claude Code, Cursor, Git/GitHub, Docker, FastAPI, Next.js, React, Kaggle